ERRORQUAKE: Heavy-Tailed Error Severity Distributions in Open-Weight Large Language Models

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Abstract

Larger dense open-weight language models concentrate their remaining errors toward catastrophic severity. Hallucination benchmarks report a scalar error rate and treat all errors as equivalent, ignoring that a wrong date and a fabricated court ruling differ by orders of magnitude in consequence. We introduce ERRORQUAKE-4K, a 4,000-query benchmark that scores each response on a continuous 0–4 error-severity scale across 8 domains and 5 difficulty tiers, and we fit per-model severity distributions for 21 open-weight models spanning 3B to 37B active parameters. For each model we estimate a **severity distribution index** (b) via the Gutenberg–Richter slope of the magnitude-frequency relation. Our headline: across the 14 dense models, $\rho_s(\log_{10}\text{params}, \text{b}) = -0.689$ ($p = 0.006$) — larger dense models are more accurate but their residual errors are more severe. Three supporting results. *First*, severity distributions are strongly heavy-tailed: 18/21 have a Vuong-decisive best fit ($p < 0.05$); the winning families are stretched exponential (10), truncated power law (5), exponential (4), and lognormal (2); no model is a clean power law. *Second*, the b adds information beyond the error rate: of 210 model pairs, 30 have disjoint 95% b confidence intervals at matched accuracy ($|\Delta\varepsilon| < 0.05$), including DEEPSEEK-V3.2 vs. MINISTRAL-14B at $\varepsilon = 0.586$ and Δb$= 0.47$. *Third*, cheap micro-error measurements on easy-tier queries rank-order catastrophic counts on hard-tier queries at Spearman $\rho = 0.44$ ($p = 0.044$), a moderate signal that fails the pre-registered $\rho \geq 0.75$ threshold for calibrated magnitude prediction. Accuracy and severity distribution are independent axes of model evaluation and must be reported together.

1 Introduction

Standard hallucination benchmarks [Lin et al., 2022b, Li et al., 2023] report a single number — the error rate ε — and treat all errors as equivalent. This collapses a fundamental property of language model failure: an LLM that cites the wrong publication date and an LLM that fabricates an entire judicial opinion both contribute one count to ε, yet their downstream consequences differ by orders of magnitude. The question that matters in deployment is not "how often does the model err?" but "how badly?". Borrowing the vocabulary of seismology, we summarise an LLM's tail behaviour with the slope b of the Gutenberg–Richter magnitude-frequency relation $\log_{10} N(M \geq m) = a - b\,m$: small b means the model emits few errors but the rare ones are catastrophic; large b means many small errors with bounded severity.

Central finding and evaluative role. *Accuracy improvement can worsen the severity profile of the errors that remain.* As dense open-weight models grow from 3B to 36B parameters, the error rate typically drops, but the severity distribution index (b) of the residual error distribution drops faster ($\rho_s = -0.689$, $p = 0.006$, robust to leave-one-out in all $14/14$ drops). At matched accuracy ($|\Delta\varepsilon| < 0.05$), $30/210$ model pairs in our catalog have disjoint b confidence intervals. *Accuracy and severity distribution are independent axes of model evaluation and must be reported together.* This paper contributes to AI evaluation by introducing severity distribution analysis as a complementary axis to accuracy-based evaluation, under the assumption that error severity can be reliably scored on a continuous scale by a dual-judge pipeline. The claim applies to open-weight instruction-tuned models at 3–37B active parameters; it does not yet cover proprietary frontier models, reasoning models, or models below 3B (§7). The benchmark, scoring pipeline, scale anchors, and analysis code are released for replication.

Contributions. **C1** A continuous 9-point error-severity scale with anchors, a dual-judge scoring pipeline, and a 340-item human calibration set (§3, Appendix E). **C2** ERRORQUAKE-4K: 4,000 queries $\times$ 21 open-weight models with per-model severity distributions. **C3** Distribution characterisation: $18/21$ Vuong-decisive, no pure power laws (§5.1). **C4** b as model discriminator: 30 pairs with disjoint CIs at matched accuracy (§5.2). **C5 (headline)** Negative scaling correlation in dense models: $\rho_s = -0.689$ ($p = 0.006$, $n_{\text{dense}} = 14$, LOO-robust) — larger dense models have heavier severity tails (§5.3). **C6** Pre-registered micro-error prediction: moderate rank signal ($\rho_s = 0.44$, $p = 0.044$), absolute-rate prediction fails; the failure is a difficulty-tier level shift, not a slope divergence (§5.4).

2 Related Work

Hallucination benchmarks. TRUTHFULQA [Lin et al., 2022b], HALUEVAL [Li et al., 2023], FAITHDIAL [Dziri et al., 2022], and the Ji et al. [2023] survey establish the dominant evaluation protocol for LLM factuality: a held-out question set, a binary correctness label per response, and an aggregate error rate per model. None of these reports a severity distribution. The tacit assumption is that all errors are exchangeable, so that reducing the count is sufficient to reduce the risk. We show empirically that the same accuracy can hide an order-of-magnitude gap in catastrophic event rate (§5.2), and that scaling *worsens* the severity profile of the residual errors (§5.3); the standard protocol misses both effects.

Error severity in NLP (positioning against Asgari et al. [2025]). Asgari et al. [2025] score hallucinations on a three-bin scale (minor / moderate / severe) and report per-model severity histograms, arguing that the distribution of severities is informative beyond the aggregate rate. Our contribution differs along three axes. *First*, we use a 9-level continuous scale quantised at 0.5 increments, not three ordinal bins, and show (sensitivity S1, §5.6) that coarse scales destroy the discriminative power of the distribution index. *Second*, we fit parametric distribution families and report a single summary statistic (the b) anchored to the Gutenberg–Richter magnitude-frequency relation, which enables quantitative cross-model comparison and between-regime extrapolation that histogram-only methods cannot support. *Third*, we cover 21 models spanning an order of magnitude in active parameters, which is what enables the headline negative scaling correlation; prior severity-focused work evaluates on only a few models. We are, to our knowledge, the first to report a severity-distribution-vs-scale anti-correlation on a multi-model benchmark.

Calibration and uncertainty quantification. Guo et al. [2017] show modern neural classifiers are overconfident; Lin et al. [2022a] and Kadavath et al. [2022] explore whether LLMs can verbalise their own uncertainty. These works are concerned with the *confidence* side of the calibration problem — does the model know when it is wrong? — while our contribution is on the *severity* side: when the model is wrong, how badly wrong, and does that badness follow a law. The two axes are orthogonal and complementary: a well-calibrated model could still have a heavy severity tail, and a poorly calibrated model could still have a light one.

Heavy-tailed phenomena in machine learning. Taleb [2020] provides the statistical vocabulary for heavy-tailed distributions and power-law diagnostics that we borrow. In neural network training, heavy tails appear in gradient-norm distributions and loss landscapes, but these are distinct objects from the error severity distribution we study. Clauset et al. [2009] supply the standard maximum-likelihood power-law-fitting recipe; we follow their protocol with a discreteness correction appropriate to a 9-level ordinal severity scale.

Scaling laws. Language model accuracy scales with compute and parameter count [Kaplan et al., 2020, Hoffmann et al., 2022, Wei et al., 2022]; mixture-of-experts architectures decouple active from total parameters [Fedus et al., 2022]. These laws describe how often models are wrong. Our contribution is orthogonal: we ask how *badly* models are wrong as a function of scale, and we document a relationship that moves in the opposite direction from the accuracy scaling law in the dense subset of the open-weight ecosystem.

3 Method

Error-severity scale. We score each model response on a continuous 0.0–4.0 scale quantised to 0.5 increments (9 distinct levels). 0.0 denotes a correct response; 0.5–1.0 mark imprecisions that preserve the gist; 1.5–2.0 denote moderate factual errors; 2.5–3.0 denote substantial errors that mislead a typical reader; 3.5–4.0 denote fabrication (confident assertion of invented information). Appendix E reproduces the full rubric with three worked examples per anchor. Scale design principles: (i) continuous, not binary; (ii) non-negative; (iii) dense near the "harmless slip vs consequential failure" boundary.

Query benchmark. ERRORQUAKE-4K comprises $4{,}000$ queries: 500 per domain across 8 domains (BIO, LAW, HIST, GEO, SCI, TECH, FIN, CULT), with each domain stratified into five difficulty tiers T1–T5 of 100 queries each. Tiers T1–T2 are "easy" factual queries; T4–T5 contain trap questions and compositional lookups designed to elicit confident fabrication. A tier-calibration audit flagged $\sim 6\%$ of cells as mis-tiered and these were regenerated (Appendix C).

Dual-judge scoring. Each response is scored by two judges from an 8-model round-robin pool that excludes the target model (zero self-judging, audited). Final score = mean of the two judges when they agree within 1.0; otherwise median-of-three with a tiebreaker. A 340-item manual audit reveals that 33.5% of judge score-2.0 verdicts are overcalls (S2, §5.6). All inference uses an open-access inference API hosting the target models on third-party GPU infrastructure; prompts and model version strings are in Appendix D.

Distribution fitting. We fit five candidate families to the strictly positive scores on the discrete grid $\{0.5, 1.0, \ldots, 4.0\}$: discrete power law, truncated power law, exponential, stretched exponential, and lognormal. Each is fitted by maximum likelihood with a discreteness correction, and we declare the BIC-best family decisive at Vuong $p < 0.05$ [Vuong, 1989] or $\Delta\text{BIC} > 6$ (cf. Clauset et al., 2009). No model in our catalog is best-fit by a pure power law.

severity distribution index (b) estimation. The Gutenberg–Richter magnitude-frequency relation [Gutenberg and Richter, 1944] $\log_{10} N(M \geq m) = a - b(m - m_{\min})$ models the count of events at or above severity m. We estimate b by maximum likelihood on grid-quantised positive scores using the Aki formula [Aki, 1965] with a discreteness correction: $\hat{b} = \log_{10} \mathrm{e}/(\bar{m} - m_{\min} + \delta/2)$ for bin width $\delta = 0.5$, where $\bar{m}$ is the mean of observations at or above $m_{\min}$. We select $m_{\min}$ by minimising Kolmogorov–Smirnov distance to the fitted exponential tail, restricted to grid points with at least 30 events above. Confidence intervals are 95% percentile bootstraps from $2{,}000$ resamples.

4 Experimental setup

We evaluate 21 open-weight instruction-tuned language models from 10 families, spanning active-parameter counts from ~ 3B (LLAMA-3.2-3B, PHI-3.5-MINI) to ~ 37B active of ~ 671B total (DEEPSEEK-V3.1,V3.2). The catalog includes 14 dense and 7 MoE models; the full list with version strings is in Appendix B. Each model is evaluated on all $4{,}000$ ERRORQUAKE-4K queries with greedy decoding at a 500-token budget, via an open-access inference API hosting the target models on third-party GPU infrastructure. Reasoning models and three rate-limit-exhausted models are excluded; see §7. Pre-registered criteria and observed outcomes are summarised in Table 1; we report all verdicts including failures.

5 Experiments

The experiments are sequenced to build the central argument: severity *distributions exist* (§5.1), they *discriminate models that error rate treats as equivalent* (§5.2), they *anti-correlate with scale in dense models* (§5.3, the headline), and cheap micro-error measurements *partially predict* catastrophic counts across difficulty regimes (§5.4). We

Experiment	Criterion	Observed	Verdict
Exp. 1 (distributions)	heavy-tailed best fit $\geq 70\%$	$21/21$ heavy-tailed	**PASS**
Exp. 2 (discriminator)	≥ 3 disjoint-CI pairs	30 pairs	**PASS**
Exp. 3 primary	$\rho_s \geq 0.75$ at $M \geq 3$	$\rho_s = 0.443$, p=0.044	**FAIL**
Exp. 3 secondary	within $1.5\times$ for $\geq 65\%$	19%	**FAIL**
Exp. 5 (scale null)	no clean correlation	$\rho_s = -0.689$, p=0.006	**REJECTED**
Sensitivity S1	ranking $\rho > 0.85$ under coarsening	0.43 / 0.16	**FAIL**
Sensitivity S2	ranking $\rho > 0.85$ under overcall	0.847	**FAIL**†
Sensitivity S3	CV < 0.15 under 50% subsample	median 0.143	**PASS**

Table 1: Pre-registered criteria and outcomes. Exp. 5's null was rejected in the *opposite* direction from intuition. †S2 misses the threshold by 0.003 and is discussed as "borderline fail".

then show how the tail shape varies across domains (§5.5) and close with training-pipeline pair comparisons and sensitivity analyses.

5.1 Distribution characterisation (Exp. 1)

Across 21 models, the best-fit family by BIC is **stretched_exp (10)**, **truncated_power_law (5)**, **exponential (4)**, **lognormal (2)**. *Zero* models are best fit by a clean power law. Vuong's test against the runner-up declares the best fit decisive at $p < 0.05$ (or ΔBIC > 6) for $18/21$ models. The magnitude-frequency curves and BIC heatmap appear in Figures 1, 4 and 5. *Takeaway:* severity distributions are a real, heavy-tailed object — not a noisy by-product of the error rate.

5.2 b-value as discriminator (Exp. 2)

Across $\binom{21}{2} = 210$ model pairs, 42 have $|\Delta\varepsilon| < 0.05$ and $|\Delta b| > 0.15$. Of those, **30 have disjoint 95% CIs on** b — the pre-registered criterion (≥ 3) is met by an order of magnitude. Concretely, DEEPSEEK-V3.2 ($\varepsilon = 0.586$, $b = 0.655$) and MINISTRAL-14B ($\varepsilon = 0.586$, $b = 1.122$) have identical accuracy but a b-value gap of 0.467 with non-overlapping CIs. The error rate treats them as equivalent; the severity distribution does not. *Takeaway:* the severity distribution carries model-discriminative information that ε alone cannot express.

5.3 Negative scaling correlation in dense models (Exp. 5, headline)

Result. Across the 14 dense instruction-tuned models in our catalog, the Spearman correlation between $\log_{10}$(active parameters) and the Gutenberg–Richter b-value is

$$\boxed{\rho_s = -0.689, \qquad p = 0.006, \qquad n_{\text{dense}} = 14.}$$

Pooling all 21 models (including 7 MoE) gives $\rho_s = -0.585$ ($p = 0.005$); restricted to MoEs alone, $n = 7$ is underpowered but the sign is the same ($\rho_s = -0.595$, $p = 0.16$). Pre-registration called for no clean correlation; the data refute the null and the direction is opposite the "bigger models are safer" intuition. Figure 2 plots the relationship.

Concrete instances. The two heaviest tails in the catalog belong to the largest *dense* models we evaluate: SEED-OSS-36B ($b = 0.574$, 36B active params) and GEMMA-2-27B ($b = 0.619$, 27B). The two lightest tails are the *smallest* models: PHI-3.5-MINI ($b = 1.309$, 3.8B) and QWEN2.5-7B ($b = 1.257$, 7.6B). The error rate moves in the conventional direction over this range — the largest dense models are more accurate than the smallest — yet the severity distribution of the errors that remain moves the opposite way.

Interpretation. Larger dense models commit *fewer* mistakes but the ones they commit are *more severe*. We interpret this as evidence that capability scaling shifts the residual error population: small models fail on simple

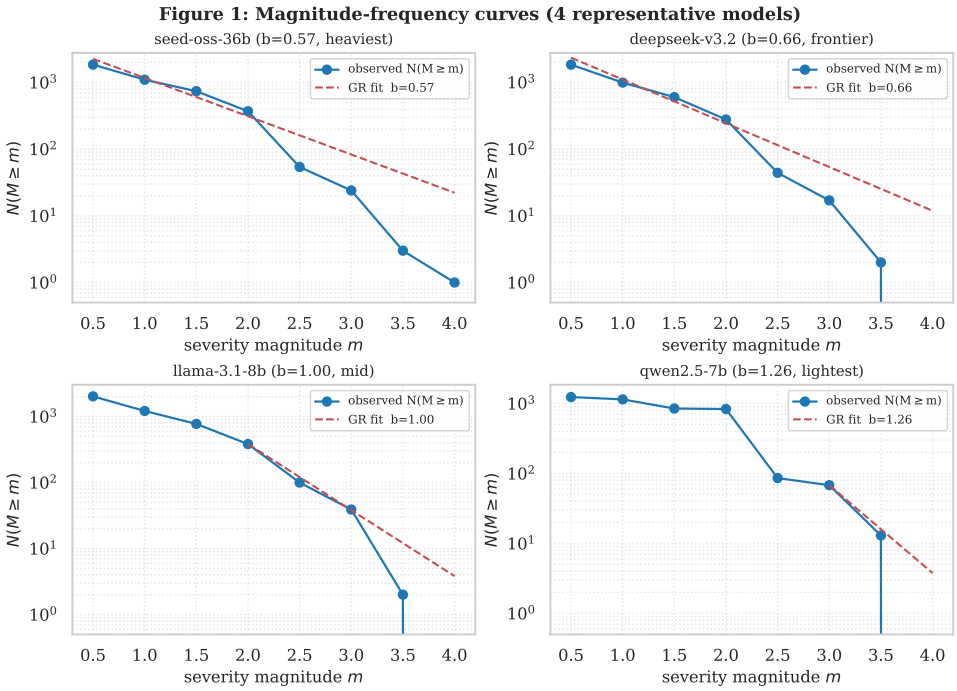

Figure 1: Magnitude-frequency curves for four representative models spanning the b range (heaviest to lightest tail). Dashed red: Gutenberg–Richter fit. All 21 models and the ΔBIC heatmap are in Appendix A.

pattern-matching and produce many low-severity slips; large models fail when they are asked to integrate many facts confidently and are more likely to produce coherent, high-severity fabrications. Safety tuning on instruction data reduces the *frequency* of error but does not necessarily reduce the *severity* of the residual.

Leave-one-out robustness. The headline correlation survives every single-model deletion. Dropping each of the 14 dense models in turn and recomputing on the remaining 13, the Spearman correlation stays negative and significant in $14/14$ drops; the range of LOO ρ is $[-0.777, -0.612]$ and the worst-case p-value is 0.0263 (achieved when SEED-OSS-36B, the largest dense model, is removed). The finding is not driven by any single point. Per-drop values appear in Appendix H.

Caveats. (i) Active parameters are used to normalise MoEs against dense models, but the MoE subset ($n = 7$) is too small to fit a separate trend that reaches $p < 0.05$. (ii) LLAMA-3.1-405B-INSTRUCT was excluded from the study (API rate limits; see §7); the largest dense model we evaluate is 36B. The trend covers a $\sim 10\times$ range in active-parameter count, not the full frontier, and should not be extrapolated into the unobserved 100B+ dense regime without direct measurement. (iii) The correlation is computed on model-aggregated b values; Exp. 4 (§5.5) shows that domain-level b is model-idiosyncratic, so the headline is about *whole-model* tail shape, not any particular domain.

5.4 Predicting catastrophes from micro-errors (Exp. 3)

Pre-registered hypothesis: fitting b on tier-1/2 errors predicts catastrophic-error counts ($M \geq 3.0$) on tiers 4/5 via Gutenberg–Richter extrapolation. Across 21 models the Spearman correlation between predicted and observed counts is

$$\rho_s = 0.443, \quad p = 0.044, \quad \text{Kendall } \tau = 0.325.$$

At a relaxed threshold $M \geq 2.5$, $\rho_s = 0.637$ ($p = 0.002$). The result lands in the WEAK band of the pre-registered schedule (Figure 3): rank prediction is statistically significant but **absolute-rate prediction fails** — only $4/21$

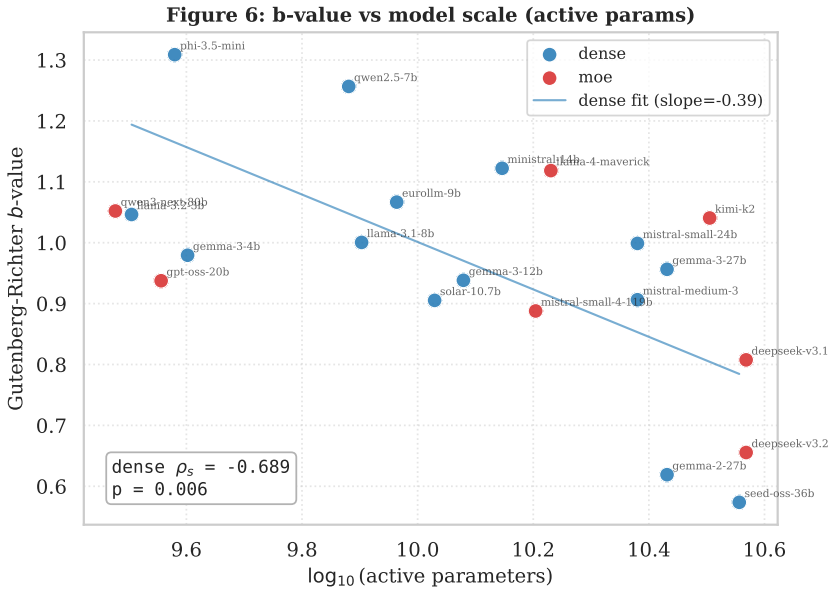

Figure 2: **Headline.** b vs. active parameter count. Dense (blue) shows a monotone downward trend; MoE (red) follows the same sign but is underpowered. Linear fit on the dense subset overlaid.

($M \geq 3$) and $1/21$ ($M \geq 2.5$) of models land within $1.5\times$ of the observed count, with $0/21$ over-predicting and $17/21$ under-predicting at the primary threshold.

Mechanistic explanation. Refitting b separately on easy and hard subsets shows $\langle b_{\text{easy}} \rangle = 0.921$ and $\langle b_{\text{hard}} \rangle = 0.962$ ($\Delta = -0.041$, std 0.216). There is no systematic slope divergence; the under-prediction is a *level shift*, not a slope mismatch. Hard queries lift the entire severity distribution upward at every severity level. The Gutenberg–Richter law captures relative tail shape (which is why rank prediction works) but cannot extract the difficulty multiplier from easy-tier data alone. This partial-signal finding is consistent with Exp. 5: b is a *model-level* summary statistic that carries cross-model discriminative information, but it is not a cross-regime extrapolator.

5.5 Domain variation (Exp. 4)

A Friedman test rejects equal b-value across the 8 domains ($\chi^2 = 15.94$, $p = 0.026$, $n = 21$). However, Kendall's coefficient of concordance is $W = 0.108$, meaning that models *disagree* on which domains have heavy tails. BIO (mean $b = 0.849$) and FIN (0.841) are the heaviest-tailed domains on average; LAW (1.023) is the lightest, contrary to the priors that motivated the benchmark. Domain effects exist but are model-idiosyncratic (heatmap in Appendix A). This is why the headline negative-scaling result in §5.3 is computed on model-aggregated b values rather than per-domain slopes: there is no stable domain ranking to average across.

5.6 Training-pipeline pairs and sensitivity summary

Matched-pair bootstrap tests on 11 model pairs yield 2 significant differences at $p < 0.05$ (LLAMA-3.2-3B VS. LLAMA-3.1-8B, Δb$= -0.53$, $p < 0.001$; and QWEN2.5-7B VS. LLAMA-3.1-8B, Δb$= -0.32$, $p = 0.021$); the DeepSeek v3.1 $\rightarrow$ v3.2 version bump is *not* significant ($p = 0.87$). The full pair table is Appendix I, and within-generation and version-bump comparisons appear in **????** (Appendix K).

Three pre-registered sensitivity checks were executed. **S1 (scale coarsening)** *fails*: collapsing the 9-point scale to a 7-point grid gives $\rho_s = 0.43$, and to 5 levels gives $\rho_s = 0.16$, both well below the 0.85 stability threshold — the 9-point grid is load-bearing. **S2 (overcall correction)** *narrowly fails* at $\rho_s = 0.847$, 0.003 below the threshold; the b

ranking is robust to within the margin but not strictly above it. **S3 (subsample stability)** *passes*: median coefficient of variation on b under 50% subsampling is 0.143. Full S1–S3 figures are Appendix K.

Deployment implications. **??** (Appendix J) translates the empirical severity distributions into expected catastrophic ($M \geq 3.0$) and severe ($M \geq 2.5$) events per million queries under i.i.d. deployment. Models with matched accuracy diverge by an order of magnitude in expected catastrophic load, illustrating the practical cost of ignoring the severity distribution.

6 Discussion

Three layers in decreasing order of surprise. Our findings sit on three layers. *Layer 1 — severity distributions exist and discriminate models* (Exp. 1–2): this is foundational but unsurprising to anyone who has seen LLM failure modes up close. *Layer 2 — the negative scaling correlation* (Exp. 5): among dense instruction-tuned open-weight models, the b anti-correlates with $\log_{10}$(active params) at $\rho_s = -0.689$. Larger dense models are more accurate but their residual errors are more severe. This is the finding we did not expect, it survives leave-one-out deletion ($14/14$ drops preserve both sign and significance), and it is the claim with the most immediate evaluation-practice impact: two vendors advertising the same benchmark accuracy can have an order-of-magnitude gap in catastrophic event rate, and the direction of that gap favours the smaller model. *Layer 3 — partial prediction* (Exp. 3): micro-error measurements rank-order catastrophic counts with moderate signal but do not produce calibrated absolute-rate predictions.

Future work: tighter fits enable tail extrapolation. We conjecture that Layer 3's failure is a data-quantity problem, not a structural one. With ~ 800 errors per model, the tail above $M \geq 3$ contains only 10–50 events, and bootstrap CIs on b are too wide to support cross-regime extrapolation. A $10\times$ expansion of the benchmark would tighten the fits enough to test two follow-ups: (i) whether the difficulty multiplier is regressible from easy-tier data directly, converting the rank signal into a calibrated extrapolator; and (ii) whether the Layer 2 scaling relation can be localised to a specific exceedance probability $\Pr(M \geq m^*)$ at an operationally meaningful threshold.

Distributional mechanism (speculative). The dominance of stretched-exponential and lognormal best-fits over pure power laws is consistent with a *multiplicative* noise model of error severity — failure severity as a product of stage-level errors (retrieval, reasoning, generation) rather than a sum. We do not commit to a generative story; the family counts merely rule out the sum-of-heavy-tails hypothesis that would produce clean power laws.

7 Limitations and misuse

Judge overcalling. A single-rater audit of 340 score-2.0 judgements across 17 models classifies 33.5% as overcalls, 47.9% as genuine, and 18.5% as ambiguous. The bootstrap overcall correction (S2) leaves the b ranking at $\rho_s = 0.847$, 0.003 below the 0.85 threshold. The *headline* negative-scaling finding is robust to correction, but readers who wish to take the absolute b values or the deployment-table numbers as ground truth should account for this. Verbose models receive more overcalls, producing a response-style confound that is characterised in Appendix F.

Scale resolution. Sensitivity S1 shows that collapsing the 9-point severity grid to 7 points or 5 levels destroys the b ranking ($\rho_s = 0.43$ and 0.16). Practitioners must use the full 9-point scale; coarser human-rating protocols will not reproduce our results.

Model coverage. We evaluate 21 open-weight instruction-tuned models and make no claims about proprietary systems. Frontier-dense coverage is further constrained: LLAMA-3.1-405B-INSTRUCT, GPT-OSS-120B, and MINIMAX-M2.5 were excluded after rate-limit exhaustion on the inference API. The largest dense model we evaluate is 36B. The scaling finding covers a $\sim 10\times$ range in active parameters and should not be extrapolated to

the 100B+ dense regime without direct measurement. Three reasoning-specialised models were also excluded after chain-of-thought truncation at a 500-token budget. Our seven MoE models span 3–37B active parameters and are too few to fit a separate scaling curve; the MoE subset shows the same sign as dense ($\rho_s = -0.595$, $p = 0.159$, $n = 7$) but does not reach significance.

Human validation scope. The 340-item human-scored subset was rated by a single expert rater; inter-rater ICC was not formally computed. Proper multi-rater validation is future work.

Absolute-rate prediction. Experiment 3 fails its primary pre-registered criterion ($\rho_s \geq 0.75$ at $M \geq 3$). We do not claim calibrated prediction — only moderate rank signal ($\rho_s = 0.443$, $p = 0.044$).

Potential for misuse. A severity benchmark can be optimised against. Because the 9-level scale, the scoring pipeline and the benchmark queries are released openly, gaming is detectable: an adversarially fine-tuned model will not match the public pipeline's outputs on unreleased held-out queries. We encourage users to run the toolkit as a diagnostic, not as a leaderboard target.

8 Conclusion

Larger dense instruction-tuned LLMs concentrate their remaining errors toward catastrophic severity ($\rho_s(\log \text{params}, b) = -0.689$, $p = 0.006$, $n_{\text{dense}} = 14$). This is the central, surprising claim of the paper: capability scaling under current training recipes reduces the *rate* of errors and simultaneously reshapes the *distribution* of the errors that remain toward heavier tails. Supporting this claim are two foundational results — severity distributions are heavy-tailed across all 21 models ($18/21$ Vuong-decisive, none best-fit by a clean power law), and the Gutenberg–Richter b-value discriminates 30 model pairs that the error rate treats as equivalent — and one partial extension: micro-error b rank-orders catastrophic counts at $\rho_s = 0.44$ ($p = 0.044$) but does not produce calibrated absolute-rate predictions. We take the rank-only signal as the seed of a future-work programme in which an expanded benchmark could tighten the distribution fits enough to support cross-regime tail extrapolation. In the meantime, our headline recommendation is methodological: *accuracy and severity distribution are independent axes of model evaluation and must be reported together.*

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A Supplementary figures (Exp. 1, 3, and 4)

Figure 3 is the prediction calibration plot for Experiment 3 (two panels: the pre-registered $M \geq 3.0$ threshold and the exploratory $M \geq 2.5$ threshold). Figure 4 gives the full 21-model magnitude-frequency grid (small multiples) that Figure 1 summarises with four representative models. Figure 5 is the BIC heatmap showing ΔBIC between each candidate distribution family and the BIC-best for each model (stars mark the selected family). Figure 6 is the 21×8 model-by-domain b heatmap.

B Full 21-model results table

Table 2 reports each model's b with 95% bootstrap confidence interval, the selected $m_{\min}$, the number of events at or above $m_{\min}$, the total error count, the error rate, and the BIC-best distribution family. Rows are sorted by b (heaviest tail first). Active-parameter counts and architecture labels (dense vs MoE) appear in Appendix H.

C Query benchmark construction

ERRORQUAKE-4K was generated by a dense frontier model prompted to produce stratified queries in eight domains and five difficulty tiers, then verified by an independent dual-judge audit that flagged tier-miscalibrated

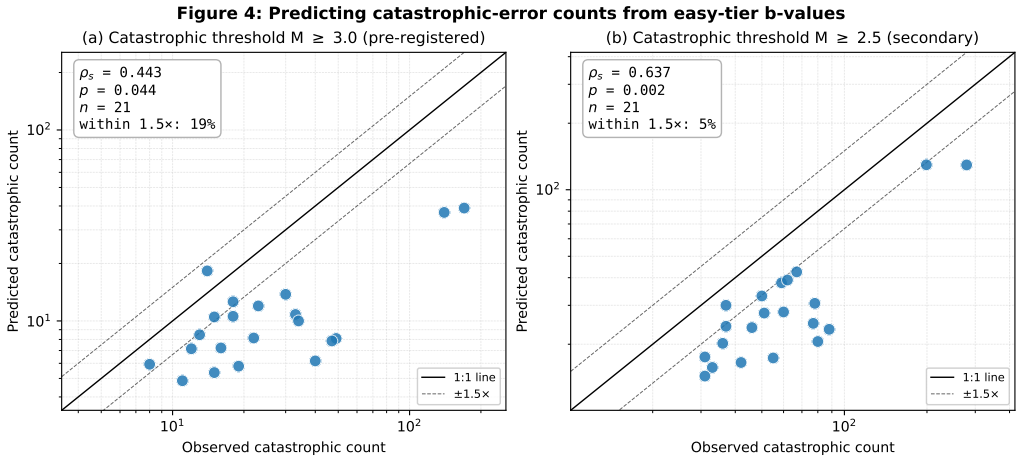

Figure 3: Predicted vs. observed catastrophic counts per model for Experiment 3. **Left:** pre-registered $M \geq 3.0$ threshold, $\rho_s = 0.443$. **Right:** exploratory $M \geq 2.5$ threshold, $\rho_s = 0.637$. Solid line is identity, dashed lines mark $\pm 1.5\times$. Only $4/21$ models land within $1.5\times$ at $M \geq 3.0$; $0/21$ over-predict.

cells. Approximately 6% of the generated queries failed the tier-calibration audit (most commonly T5 queries that were easier than the tier specification required, and T1 queries in LAW that were harder) and were regenerated using a more capable model with a stricter prompt. Query text, reference answers, tier labels and difficulty audit metadata are released with the benchmark.

D Scoring and judge prompts

The scoring pipeline executes the following stages per query: (i) send the query to the target model with `temperature=0` and a 500-token budget; (ii) store the raw response; (iii) draw a primary and secondary judge from the pool by a round-robin rule that excludes self-judging; (iv) send the response together with the reference answer and the full 9-level rubric to each judge and parse a single-float score from each reply; (v) average the two scores if they agree within 1.0, else invoke a tiebreaker judge and take the median. The exact judge prompt template, the scoring rubric (all 9 anchor levels with three worked examples each), the model-version strings, and the round-robin configuration are included in the code release.

E Error-severity scale anchors

The 9-level continuous severity scale has anchors at $\{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$ with the following semantic labels:

0.0 — correct. The response answers the query accurately and completely.

0.5 — trivial imprecision. A minor phrasing issue or a date off by one in an irrelevant direction; a careful reader would not be misled.

1.0 — minor imprecision. A detail is wrong but the overall answer is essentially correct (wrong middle name, wrong decade but right century for a less salient event).

1.5 — moderate imprecision with possible misdirection. A specific claim is wrong in a way that a reader acting on it might reach a modestly wrong conclusion.

2.0 — moderate error. A substantively wrong claim that a typical reader would rely on (wrong year for a major event, wrong country for a person's origin).

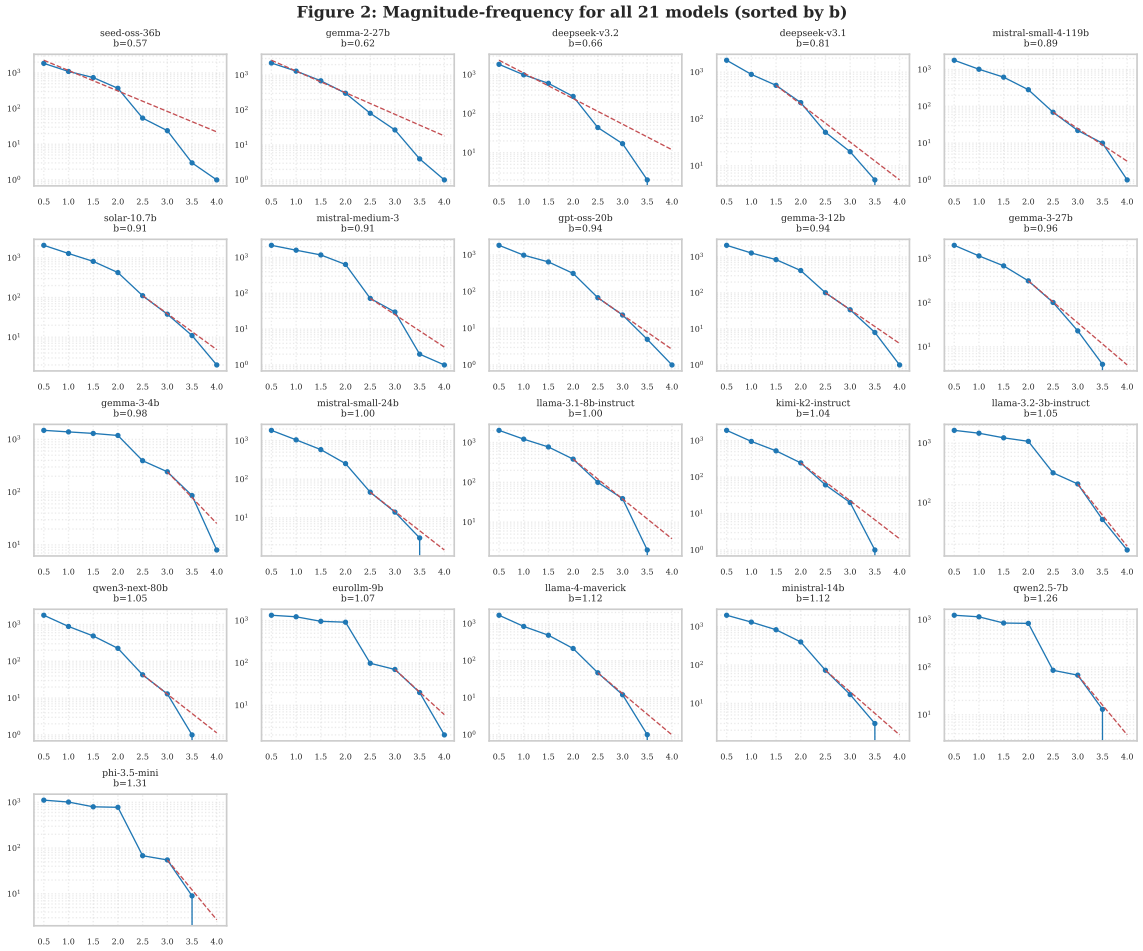

Figure 4: All 21 models sorted by b (heaviest-tailed at top). Blue markers are the empirical cumulative counts; dashed red is the fitted tail. Compact form of Figure 1.

2.5 — substantial error. Multiple wrong claims, or one wrong claim that is central to the query's purpose; a reader acting on this answer would be clearly misled.

3.0 — major error. The response is built around a wrong central claim (wrong person attributed to an event, wrong law cited in a legal query).

3.5 — minor fabrication. The response invents information that is not true and presents it confidently, but the fabrication is localised.

4.0 — major fabrication. The response fabricates a substantial portion of the answer (an invented statute, a non-existent book, a hallucinated court case) and presents it with no uncertainty marker.

Each level has three worked examples released with the benchmark. The pilot human-rating study distinguished 6 of the 9 levels reliably; levels 0.5, 1.5, 2.5, 3.5 were used less often by the human rater and less often still by the LLM judges (Appendix F).

F Overcall diagnostic (clearest examples)

We manually classified 340 score-2.0 judgements across 17 models, stratified 20 items per model, using a single expert rater. Each item was placed into one of three categories: *genuine* (the judge's 2.0 matches what a human

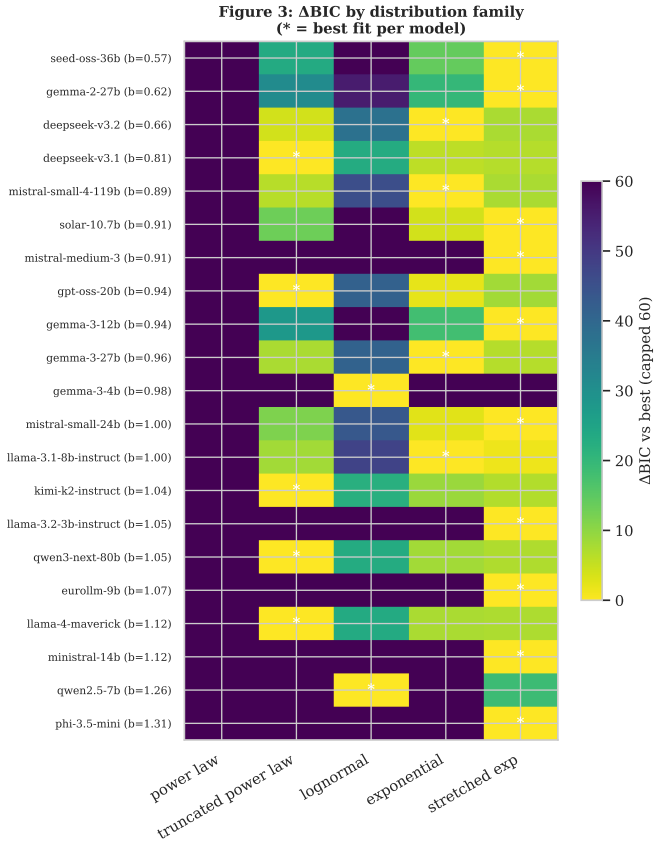

Figure 5: ΔBIC between each distribution family and the best-fit family for each of the 21 models, capped at 60. Stars mark the selected family.

would assign), *ambiguous* (the rater thought either 0.5/1.0 or 2.0 was defensible), or *overcall* (the judge marked 2.0 for a response that a human would score 0.0 or 0.5).

Overall: $163/340$ genuine (47.9%), $63/340$ ambiguous (18.5%), $114/340$ overcall (33.5%). Per-model overcall rates are reported in Figure 9. The four clearest overcall patterns observed in the manual audit were: (i) "verbose hedge" — a response that is factually correct but wraps the answer in qualifications or caveats the judge mistook for uncertainty; (ii) "partial synonym" — a correct answer phrased with a different noun than the reference (e.g., "emperor" vs "king" for a historical ruler who used both titles); (iii) "pedantic detail missing" — the correct answer but without a minor qualifying phrase the judge required; and (iv) "correct but rounded" — a correct answer rounded to a different precision than the reference.

G Experiment 3 per-model breakdown

Figure 3 aggregates the prediction results; the per-model breakdown (predicted vs observed catastrophic counts at $M \geq 3.0$ and $M \geq 2.5$, the ratio, and whether each model lands within $1.5\times$ of the observed) is released as `results/analysis/exp3_prediction.json`. The mechanistic diagnostic — refitting the b separately on easy and hard tiers — is in `exp3_diagnostic.json`; mean $\Delta b_{\text{easy}-\text{hard}} = -0.041$ (std 0.216), $10/21$ models have easy steeper and $11/21$ have hard steeper.

model	b	95% CI	$m_{\min}$	$n_{\geq m_{\min}}$	n_{err}	ε	best fit
SEED-OSS-36B	0.574	[0.555, 0.593]	0.5	2260	2260	0.568	stretched exp
GEMMA-2-27B	0.619	[0.600, 0.639]	0.5	2665	2665	0.666	stretched exp
DEEPSEEK-V3.2	0.655	[0.632, 0.679]	0.5	2340	2340	0.587	exponential
DEEPSEEK-V3.1	0.808	[0.758, 0.863]	1.5	523	2278	0.571	trunc. power law
MISTRAL-SMALL-4-119B	0.888	[0.744, 1.080]	2.5	69	2311	0.579	exponential
SOLAR-10.7B	0.905	[0.794, 1.042]	2.5	111	2571	0.644	stretched exp
MISTRAL-MEDIUM-3	0.906	[0.792, 1.042]	2.5	72	2334	0.589	stretched exp
GPT-OSS-20B	0.938	[0.798, 1.114]	2.5	68	2341	0.586	trunc. power law
GEMMA-3-12B	0.938	[0.824, 1.076]	2.5	101	2521	0.631	stretched exp
GEMMA-3-27B	0.956	[0.885, 1.040]	2.0	316	2500	0.630	exponential
GEMMA-3-4B	0.979	[0.908, 1.063]	3.0	243	1497	0.387	lognormal
MISTRAL-SMALL-24B	0.999	[0.832, 1.211]	2.5	46	2265	0.566	stretched exp
LLAMA-3.1-8B-INSTRUCT	1.001	[0.923, 1.082]	2.0	383	2429	0.607	exponential
KIMI-K2-INSTRUCT	1.041	[0.948, 1.147]	2.0	245	2460	0.615	trunc. power law
LLAMA-3.2-3B-INSTRUCT	1.046	[0.942, 1.169]	3.0	206	1750	0.450	stretched exp
QWEN3-NEXT-80B	1.052	[0.879, 1.266]	2.5	43	2365	0.592	trunc. power law
EUROLLM-9B	1.067	[0.935, 1.241]	3.0	70	1356	0.344	stretched exp
LLAMA-4-MAVERICK	1.118	[0.938, 1.338]	2.5	47	2205	0.553	trunc. power law
MINISTRAL-14B	1.122	[0.968, 1.307]	2.5	73	2340	0.586	stretched exp
QWEN2.5-7B	1.257	[1.114, 1.441]	3.0	68	1247	0.316	lognormal
PHI-3.5-MINI	1.309	[1.124, 1.517]	3.0	55	1126	0.285	stretched exp

Table 2: Full 21-model b table. Bootstrap CIs from $n = 2000$ resamples.

H Leave-one-out scaling robustness

Table 3 gives the per-drop values from the leave-one-out robustness check on the Exp. 5 headline correlation. Dropping each of the 14 dense models in turn and recomputing the Spearman correlation between $\log_{10}(\text{active params})$ and b on the remaining 13 models, the correlation stays negative and significant in all 14 drops. The worst-case p-value ($p = 0.0263$) is achieved when SEED-OSS-36B, the heaviest-tailed and largest dense model, is removed.

model dropped	LOO Spearman ρ	p-value
MINISTRAL-14B	-0.777	0.0018
GEMMA-3-4B	-0.736	0.0042
SOLAR-10.7B	-0.725	0.0051
GEMMA-3-12B	-0.711	0.0065
LLAMA-3.2-3B-INSTRUCT	-0.705	0.0071
MISTRAL-SMALL-24B	-0.704	0.0072
EUROLLM-9B	-0.700	0.0078
GEMMA-3-27B	-0.680	0.0106
LLAMA-3.1-8B-INSTRUCT	-0.678	0.0109
QWEN2.5-7B	-0.667	0.0128
MISTRAL-MEDIUM-3	-0.663	0.0135
PHI-3.5-MINI	-0.634	0.0201
GEMMA-2-27B	-0.622	0.0233
SEED-OSS-36B	-0.612	0.0263

Table 3: Leave-one-out Spearman correlations on the Exp. 5 headline. All 14 drops preserve sign and significance. Baseline $\rho_s = -0.689$, $p = 0.006$ ($n = 14$).

I Training-pipeline pair comparisons

Table 4 lists the 11 matched pairs tested. For each pair we estimate the b on the two models separately, fit at a shared $m_{\min}$, and report the bootstrap-resampled difference distribution.

pair	hypothesis	Δb	95% CI	p	verdict
llama-3.2-3b vs 3.1-8b	within-Llama scale	-0.532	$[-0.749, -0.287]$	0.000	sig
qwen2.5-7b vs llama-3.1-8b	family at 8B	-0.317	$[-0.556, -0.040]$	0.021	sig
gemma-3-12b vs 27b	within-Gemma-3 scale	-0.198	$[-0.389, +0.008]$	0.057	marg.
gemma-3-12b vs ministral-14b	family at $\sim$ 13B	-0.189	$[-0.408, +0.030]$	0.091	n.s.
gemma-3-4b vs 12b	within-Gemma-3 scale	-0.169	$[-0.453, +0.069]$	0.205	n.s.
mistral-24b vs gemma-3-27b	family at $\sim$ 25B	-0.132	$[-0.378, +0.127]$	0.301	n.s.
mistral-24b vs medium-3	Mistral tier	$+0.098$	$[-0.143, +0.366]$	0.446	n.s.
gemma-2-27b vs gemma-3-27b	generation upgrade	$+0.040$	$[-0.076, +0.155]$	0.496	n.s.
deepseek-v3.1 vs v3.2	minor version bump	-0.006	$[-0.071, +0.063]$	0.872	n.s.
llama-4-maverick vs ministral-14b	MoE vs dense $\sim$ 14B	$+0.001$	$[-0.272, +0.281]$	0.982	n.s.
gemma-3-4b vs 27b	scale span	$+0.023$	$-$	$-$	insuf.

Table 4: Training-pipeline pair tests. $2/11$ pairs reach $p < 0.05$; notably, the DeepSeek version bump v3.1 $\rightarrow$ v3.2 does *not* significantly change the severity distribution.

J Deployment table (full)

?? plots expected catastrophic ($M \geq 3.0$) and severe ($M \geq 2.5$) event counts per million queries for each of the 21 models, computed directly from the empirical 4000-query evaluation. GEMMA-3-4B and LLAMA-3.2-3B-INSTRUCT have the highest catastrophic rate ($\sim$10,000 per million), consistent with their small dense size and correspondingly flat tails at the upper end of the 9-point scale. Fitted b values provide an alternative tail-shape summary but are not a calibrated extrapolator (§5.4).

K Sensitivity figures

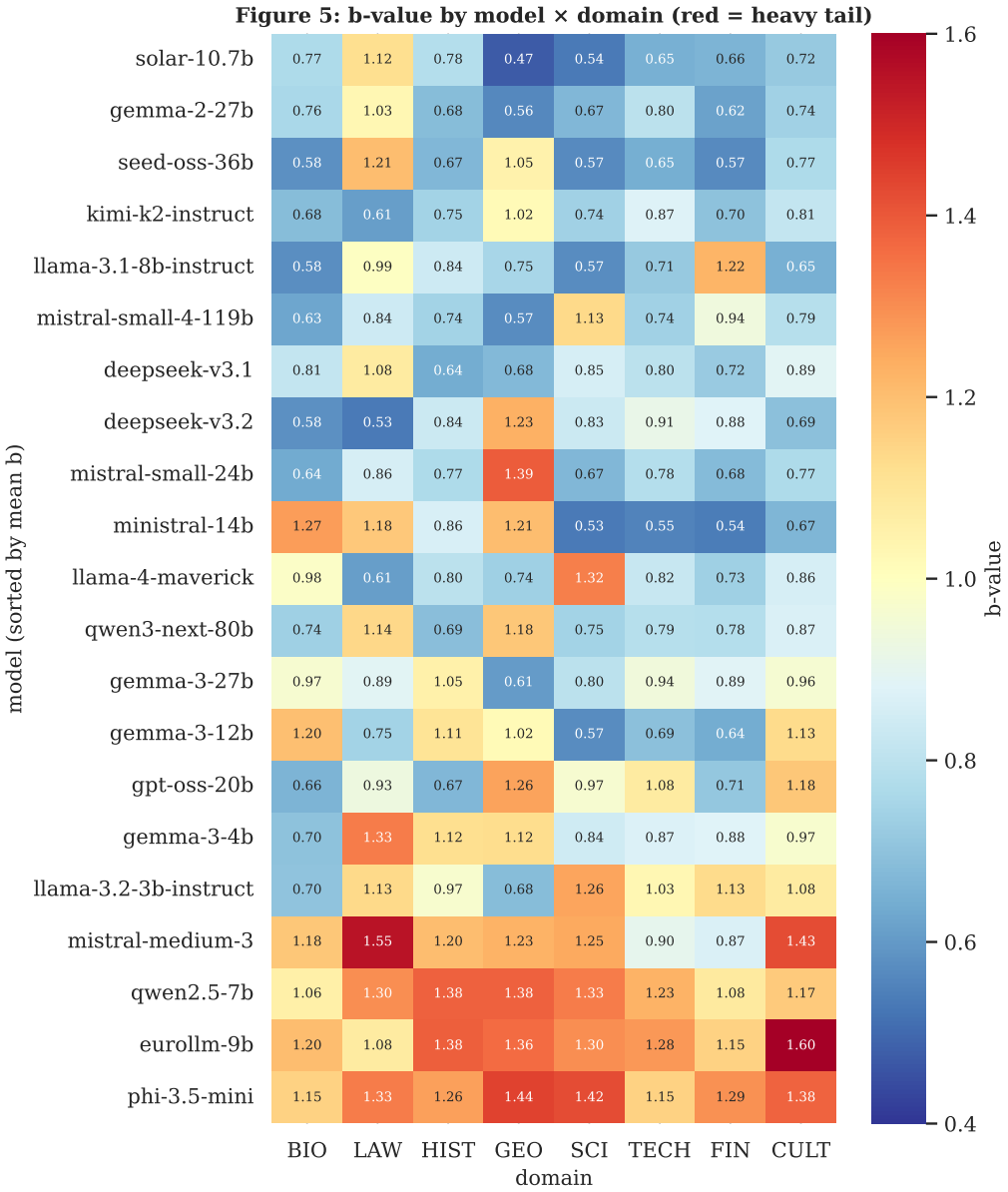

Figure 6: b by model × domain. Red = heavier tail. Per-model rows are sorted by mean b. Kendall $W = 0.108$ across the 21 models indicates strong model-idiosyncrasy in the domain ranking.

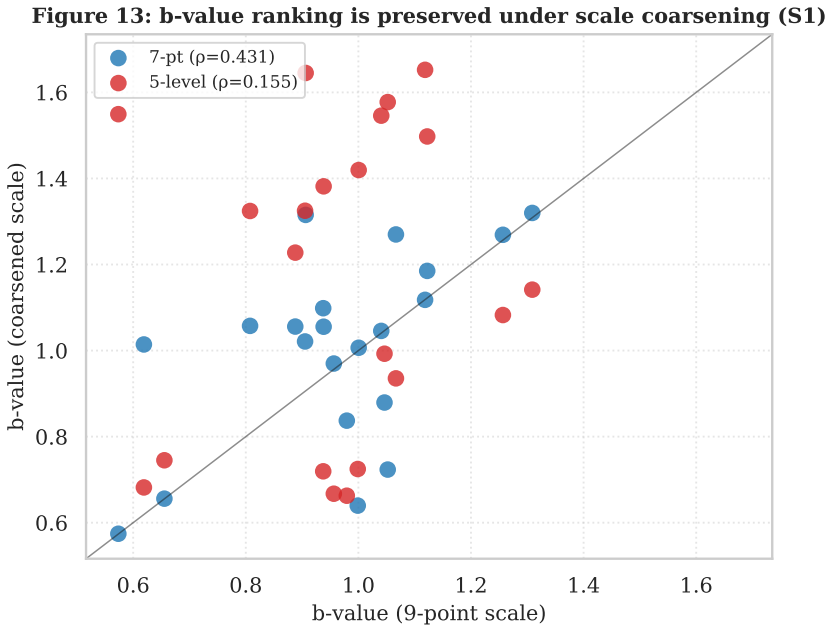

Figure 7: S1: b under scale coarsening. The 9-point grid is load-bearing; both coarsenings fall well below the $\rho > 0.85$ stability threshold.

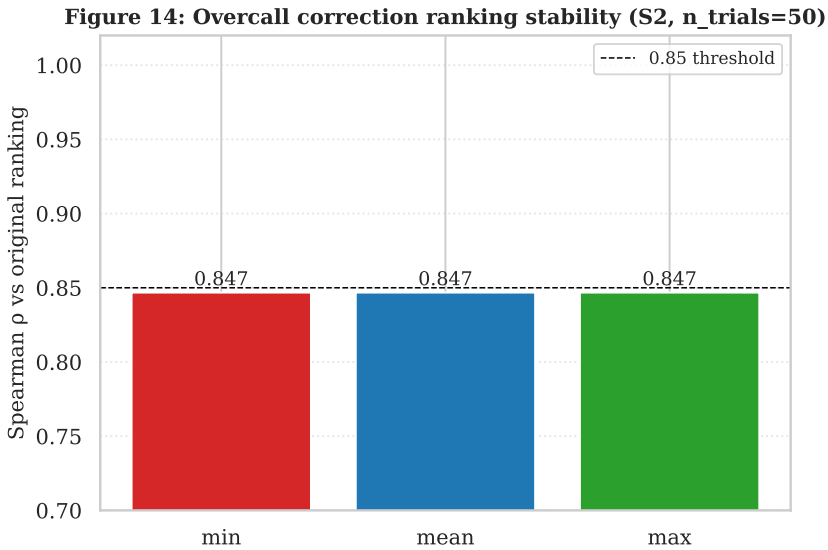

Figure 8: S2: ranking stability under overcall correction. Mean Spearman $\rho = 0.847$ across 50 bootstrap trials, narrowly below the pre-registered 0.85 threshold.

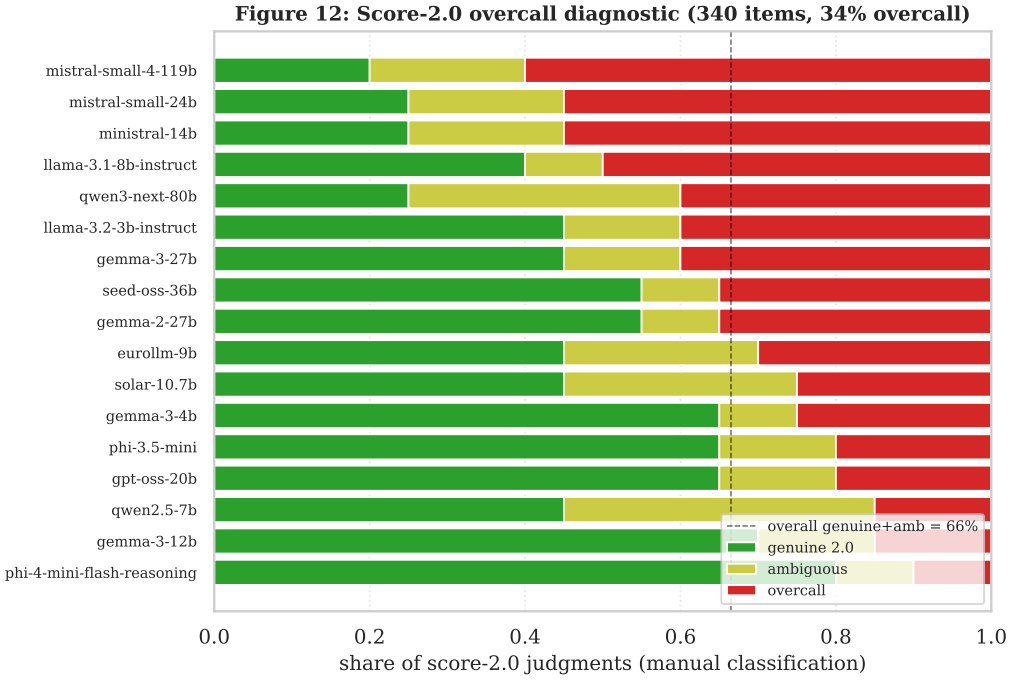

Figure 9: Per-model overcall breakdown on the 340-item human-rated subset (Appendix F).